KS3 DIGITAL TECHNOLOGY

CLOUD COMPUTING AND GREEN COMPUTING

What really happens when you save to "the cloud": cloud service models, what's inside a data centre, how cloud services scale and sometimes fail, how to keep cloud data secure, the energy and water cost of computing and AI, and practical ways to make computing greener.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 What is the cloud?
2 Inside a data centre
3 Scaling and outages
4 Securing the cloud
5 The environmental cost
6 Green computing

LESSON 1 OF 6

WHAT IS THE CLOUD?

Cloud computing means using computing services such as storage, software and processing power over the internet, from servers owned by a provider, instead of on your own device.

STARTER · 5 MINUTES

DO NOW

"The cloud is just someone else's computer." Is this true? Discuss.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Define cloud computing, describe IaaS, PaaS and SaaS, and evaluate the cloud's advantages and disadvantages

SUCCESS CRITERIA: I CAN...

- ✓ I can define cloud computing
- ✓ I can describe SaaS, PaaS and IaaS with examples
- ✓ I can evaluate advantages and disadvantages of using the cloud

"SOMEONE ELSE'S COMPUTER"

- Cloud computing: using storage, software and processing over the internet
- Runs on servers in data centres owned by cloud providers
- Big providers: Amazon Web Services, Microsoft Azure, Google Cloud
- You rent what you need instead of buying your own hardware

KEY WORDS

CLOUD COMPUTING	Using computing services such as storage, software and processing over the internet.
SERVER	A powerful computer that provides services to other computers.
SAAS	Software as a Service: finished apps used over the internet.
PAAS	Platform as a Service: a cloud platform for building and running your own apps.
IAAS	Infrastructure as a Service: renting virtual servers, storage and networks.

THREE SERVICE MODELS

- SaaS (Software as a Service): use a finished app, e.g. Google Docs, Microsoft 365, Spotify
- PaaS (Platform as a Service): a platform to build and run your own apps
- IaaS (Infrastructure as a Service): rent virtual servers, storage and networks
- Think pizza: eat out (SaaS), takeaway (PaaS), buy ingredients (IaaS)

ACTIVITY · 15 MINUTES · PAIRS

CLOUD SORT

1. Sort the 12 service cards (e.g. Gmail, a rented virtual server, Netflix, an app-hosting platform, OneDrive) into SaaS, PaaS and IaaS.

2. For three of them, write one advantage and one disadvantage of using it in the cloud.

3. Decide: should our school store everything in the cloud? Give a reason.

In your workbook, answer:

? Which model is Microsoft 365, and why?

? Should the school store everything in the cloud?

Your workbook: Lesson 1 · tick each step off as you go

PROS AND CONS

- Access from anywhere, on any device
- Easy sharing and collaboration
- No need to buy and maintain servers; pay for what you use
- But: needs internet, depends on the provider, privacy and ongoing costs

QUICK CHECK

1. What is cloud computing?

A Weather forecasting
B A type of printer
C Storing data on a USB stick
D Using computing services over the internet

2. Which is an example of SaaS?

A A rented virtual server
B Google Docs
C A motherboard
D A USB stick

3. In IaaS, what do you rent?

A A finished app
B A laptop
C Virtual servers, storage and networks
D A pizza

4. Which is a disadvantage of the cloud?

A It needs an internet connection
B Automatic backups
C Access from anywhere
D Easy sharing

QUICK CHECK: ANSWERS

1. **D: Using computing services over the internet**
   Services from remote servers.

2. **B: Google Docs**
   Finished apps.

3. **C: Virtual servers, storage and networks**
   Infrastructure.

4. **A: It needs an internet connection**
   No internet, no access.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Define cloud computing.

Look back at today's success criteria: can you tick them all?

- ✓ I can define cloud computing
- ✓ I can describe SaaS, PaaS and IaaS with examples
- ✓ I can evaluate advantages and disadvantages of using the cloud

LESSON 2 OF 6

INSIDE A DATA CENTRE

A data centre is a building full of servers, storage and networking equipment in racks.

STARTER · 5 MINUTES

DO NOW

Your laptop gets hot when you game. Imagine 100,000 computers in one building: what problems would you need to solve?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe what is inside a data centre and how it is kept cool, powered and running

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the main parts of a data centre
- ✓ I can explain why cooling and backup power are needed
- ✓ I can explain redundancy and why data centres are built where they are

WHAT'S INSIDE

- Rows of racks: tall cabinets holding many servers
- Storage systems holding enormous amounts of data
- High-speed networking connects everything to the internet
- AI data centres: thousands of GPUs for training and running AI

KEY WORDS

DATA CENTRE	A building that houses large numbers of servers, storage and networking equipment.
RACK	A tall cabinet that holds many servers.
COOLING	Removing heat produced by computers so they don't overheat.
REDUNDANCY	Having spare equipment and copies so a failure doesn't stop a service.
UPTIME	The percentage of time a service is working.

KEEPING IT RUNNING

- Power: as much electricity as a town; backup batteries and generators
- Cooling: fans, chilled air, water cooling, even liquid cooling directly on chips
- Redundancy: spare parts and copies so one failure doesn't stop the service
- Security: fences, guards, cameras, ID checks, fire suppression

WHERE AND WHY

- Near cheap, reliable (ideally renewable) electricity
- Cool climates reduce cooling costs
- Good fibre connections and low risk of floods
- Ireland hosts many data centres: they use about a fifth of its electricity

ACTIVITY · 25 MINUTES · GROUPS OF 3

BUILD A DATA CENTRE

1. Your company wants to build a data centre on the island of Ireland. Look at the four site cards (power supply, climate, flood risk, fibre, local community).
2. Score each site and choose the best one.
3. Design the cooling, power and security systems for your data centre.
4. Write one way your data centre could benefit the local community.

In your workbook, answer:

? Which site did you choose, and why?

? How could the data centre benefit the community?

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What is a rack in a data centre?

A A shelf for books
B A tall cabinet holding many servers
C A cooling fan
D A cable

2. Why do data centres need cooling?

A To keep staff comfortable only
B Because of rain
C To make ice
D Servers turn electricity into heat that can damage them

3. What is redundancy?

A Spare equipment and copies so a failure doesn't stop the service
B Deleting data
C Using one server
D Losing your job

4. What does 99.9% uptime allow per year?

A About 9 minutes
B No downtime
C About 9 hours of downtime
D About 9 days

QUICK CHECK: ANSWERS

1. **B: A tall cabinet holding many servers**
   Servers are stacked in racks.

2. **D: Servers turn electricity into heat that can damage them**
   Heat damages electronics.

3. **A: Spare equipment and copies so a failure doesn't stop the service**
   Backups that take over.

4. **C: About 9 hours of downtime**
   0.1% of 8,760 hours.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three systems needed to keep a data centre running.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the main parts of a data centre
- ✓ I can explain why cooling and backup power are needed
- ✓ I can explain redundancy and why data centres are built where they are

LESSON 3 OF 6

SCALING AND OUTAGES

Virtualisation lets one physical server run many virtual machines, and cloud services scale up and down automatically to match demand.

STARTER · 5 MINUTES

DO NOW

A concert sells tickets online at 9am and two million fans log on at once. What could go wrong?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain virtualisation and scalability, and describe causes and effects of cloud outages

SUCCESS CRITERIA: I CAN...

- ✓ I can explain virtualisation
- ✓ I can explain scalability and elasticity
- ✓ I can describe causes and impacts of cloud outages

VIRTUALISATION

- One physical server can run many virtual machines (VMs)
- Each VM acts like a separate computer with its own operating system
- Makes better use of hardware: less waste
- Containers are a lighter-weight version

KEY WORDS

VIRTUALISATION	Using software to run many virtual computers on one physical machine.
VIRTUAL MACHINE	A software-based computer that runs on a physical server.
SCALABILITY	The ability of a system to handle more work by adding resources.
LOAD BALANCER	A system that spreads requests across many servers.
OUTAGE	A period when a service stops working.

SCALING UP AND DOWN

- Scalability: handling more users by adding resources
- Elasticity: scaling automatically up and down with demand
- Load balancers share traffic between many servers
- Pay only for what you use: ideal for ticket sales, Black Friday, exam results day

WHEN THE CLOUD BREAKS

- Outages are caused by software bugs, bad updates, configuration mistakes, power cuts or cyber attacks
- Many apps rely on the same few providers, so one fault spreads
- October 2025: a fault in one AWS region took down many popular apps and websites for hours
- Resilience: spread services across regions and providers

ACTIVITY · 25 MINUTES · GROUPS OF 3

OUTAGE RESPONSE TEAM

1. Read the incident report: "Ticketing site crashed for 3 hours during a big concert sale."
2. Identify what probably went wrong (use the scaling and outage notes).
3. Write a timeline of what the team should do during the outage.
4. Write three changes to stop it happening again.

In your workbook, answer:

? What probably caused the crash?

? Write three changes to prevent it next time.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is virtualisation?

A Playing VR games
B Using paper
C Deleting servers
D Running many virtual machines on one physical server

2. What does elasticity mean in cloud computing?

A Fixed number of servers
B Stretchy cables
C Slower servers
D Scaling up and down automatically with demand

3. What does a load balancer do?

A Cools servers
B Spreads requests across many servers
C Weighs servers
D Stores passwords

4. Why can one cloud fault affect millions of people?

A Many services rely on the same providers
B Everyone uses the same password
C The internet is one computer
D It can't

QUICK CHECK: ANSWERS

1. **D: Running many virtual machines on one physical server**
Better use of hardware.

2. **D: Scaling up and down automatically with demand**
Matches demand.

3. **B: Spreads requests across many servers**
Prevents overload.

4. **A: Many services rely on the same providers**
Shared dependence.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain one benefit of virtualisation.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain virtualisation
- ✓ I can explain scalability and elasticity
- ✓ I can describe causes and impacts of cloud outages

LESSON 4 OF 6

SECURING THE CLOUD

Keeping cloud data safe is a shared responsibility: providers secure the infrastructure, but users must protect their accounts, control sharing and back up important data.

STARTER · 5 MINUTES

DO NOW

You've shared a Google Drive folder with "Anyone with the link". Who could see it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how to keep data in the cloud secure, including shared responsibility, access control and backups

SUCCESS CRITERIA: I CAN...

- ✓ I can explain the shared responsibility model
- ✓ I can describe ways to secure cloud accounts and sharing
- ✓ I can apply the 3-2-1 backup rule

SHARED RESPONSIBILITY

- The provider secures the buildings, hardware and network
- You secure your account, passwords and sharing settings
- Most cloud data leaks are caused by mistakes, e.g. files shared publicly by accident
- Security is a team effort

KEY WORDS

SHARED RESPONSIBILITY	The idea that cloud providers and users each have security jobs to do.
TWO-FACTOR AUTHENTICATION	A second check, such as a code on your phone, as well as a password.
ENCRYPTION	Scrambling data so only someone with the key can read it.
BACKUP	A copy of data kept so it can be restored if lost.
DATA SOVEREIGNTY	The idea that data is subject to the laws of the country where it is stored.

PROTECT YOUR ACCOUNT

- Strong, unique passwords (or passkeys)
- Two-factor authentication (2FA)
- Check sharing: specific people, not "anyone with the link"
- Watch for phishing emails pretending to be your cloud provider

BACKUPS AND SOVEREIGNTY

- 3-2-1 rule: 3 copies, on 2 different types of storage, 1 off-site
- Sync is not backup: deleting a file can delete it everywhere
- Encryption protects data stored and sent
- Data sovereignty: which country's laws apply to where your data is stored

ACTIVITY · 20 MINUTES · PAIRS

CLOUD SECURITY AUDIT

1. Read the profile of "Jordan's" cloud account: password used on 4 sites, no 2FA, 12 folders shared with "anyone with the link", no backups, phone is the only device.
2. Identify at least five risks.
3. Write a step-by-step plan to secure Jordan's account, in priority order.
4. Design a 3-2-1 backup plan for Jordan's photos and schoolwork.

In your workbook, answer:

? List Jordan's risks.

? Write Jordan's 3-2-1 backup plan.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. In the shared responsibility model, who secures your password and sharing settings?

A You, the user
B The cloud provider
C The government
D Nobody

2. What is 2FA?

A Two accounts
B Two passwords written down
C A type of virus
D A second check, like a code on your phone, as well as a password

3. What does the 3-2-1 backup rule say?

A 3 days, 2 weeks, 1 month
B 3 copies, 2 types of storage, 1 off-site
C 3 passwords, 2 phones, 1 laptop
D 3 clouds only

4. Why is syncing not the same as backup?

A Backups use the internet
B Deletions and damage can sync to every copy
C They're the same
D Syncing is slower

QUICK CHECK: ANSWERS

1. **A: You, the user**
   Users protect their accounts.

2. **D: A second check, like a code on your phone, as well as a password**
   Extra layer of security.

3. **B: 3 copies, 2 types of storage, 1 off-site**
   A simple backup plan.

4. **B: Deletions and damage can sync to every copy**
   Sync copies mistakes too.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the shared responsibility model.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain the shared responsibility model
- ✓ I can describe ways to secure cloud accounts and sharing
- ✓ I can apply the 3-2-1 backup rule

LESSON 5 OF 6

THE ENVIRONMENTAL COST

Data centres use around 1-2% of the world's electricity, and AI is making demand grow quickly.

STARTER · 5 MINUTES

DO NOW

Does sending an email, streaming a video or asking an AI chatbot have an environmental cost? Rank them.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe the environmental impact of data centres, AI and devices, including energy, water and e-waste

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how data centres affect the environment
- ✓ I can explain why AI is increasing energy demand
- ✓ I can describe the impact of making and disposing of devices

ENERGY

- Data centres use roughly 1.5% of the world's electricity, and rising
- AI is a major driver: training and running large models needs huge computing power
- Forecasts suggest data centre electricity could roughly double by 2030
- If electricity comes from fossil fuels, that means carbon emissions

KEY WORDS

CARBON FOOTPRINT	The total greenhouse gases caused by a person, product or organisation.
RENEWABLE ENERGY	Energy from sources that don't run out, such as wind, solar and hydro.
E-WASTE	Thrown-away electronic devices and parts.
EMBODIED CARBON	The carbon emitted while making a product, before it is used.

WATER AND LAND

- Many data centres use water to cool servers
- Large sites can use millions of litres a day
- Concerns in places with water shortages
- Data centres need land, grid connections and new power stations

DEVICES AND E-WASTE

- Most of a phone's lifetime carbon comes from making it (embodied carbon)
- Mining metals for chips and batteries harms environments
- E-waste is one of the fastest-growing waste streams in the world
- Keeping devices longer is one of the biggest green actions

ACTIVITY · 25 MINUTES · PAIRS

FOOTPRINT DETECTIVES

1. Use the (approximate) footprint cards for digital activities: an hour of HD streaming, an email with a big attachment, 10 AI chatbot questions, generating an AI image, making a new phone.
2. Rank them from lowest to highest impact.
3. Calculate your class's weekly streaming footprint using the card values.
4. Decide which single change would make the biggest difference.

In your workbook, answer:

? Which activity has the biggest impact, and why?

? Which single change makes the biggest difference?

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. Roughly what share of world electricity do data centres use?

A About 50%
B None
C About 1-2%
D About 90%

2. Why is AI increasing data centre energy demand?

A AI runs without electricity
B It isn't
C Training and running large models needs huge computing power
D AI makes servers cooler

3. What is embodied carbon?

A Carbon in your body
B Carbon paper
C Carbon in the air
D Carbon emitted making a product

4. Why do some data centres use lots of water?

A For cooling servers
B For staff showers
C To clean the floors
D For fish tanks

QUICK CHECK: ANSWERS

1. **C: About 1-2%**
   Small share, but growing fast.

2. **C: Training and running large models needs huge computing power**
   AI is compute-hungry.

3. **D: Carbon emitted making a product**
   Before you switch it on.

4. **A: For cooling servers**
   Evaporative cooling.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three ways data centres affect the environment.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how data centres affect the environment
- ✓ I can explain why AI is increasing energy demand
- ✓ I can describe the impact of making and disposing of devices

LESSON 6 OF 6

GREEN COMPUTING

Green computing means designing, using and disposing of technology in ways that reduce its environmental impact.

STARTER · 5 MINUTES

DO NOW

Name one thing you could change about how you use technology this week to reduce its environmental impact.

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how computing can be made greener by organisations, programmers and individuals

SUCCESS CRITERIA: I CAN...

- ✓ I can describe green computing actions for data centres
- ✓ I can explain how efficient software reduces energy use
- ✓ I can make a personal green computing plan

GREENER DATA CENTRES

- Power from renewable energy
- Reuse waste heat to warm homes, offices, greenhouses and swimming pools
- Efficient cooling, e.g. using cold outside air or liquid cooling
- Carbon-aware computing: run big jobs when the grid is greenest

KEY WORDS

GREEN COMPUTING	Designing, using and disposing of technology in ways that reduce environmental harm.
HEAT REUSE	Using waste heat, for example from data centres, to warm buildings.
RIGHT TO REPAIR	Rules that make it easier for people to repair their own devices.
CARBON-AWARE COMPUTING	Running computing jobs when and where electricity is cleanest.

GREENER CODE AND AI

- Efficient algorithms do the same job with fewer steps, so less energy
- Smaller AI models can be good enough for many tasks
- Compress images and videos; don't load what users don't need
- Measure energy use, like any other bug to fix

GREENER YOU

- Keep devices longer; repair, don't replace
- Recycle e-waste properly; donate working devices
- Turn off and unplug; use power-saving modes
- Stream at lower quality on small screens; delete files you don't need

ACTIVITY · 30 MINUTES · GROUPS OF 3

GREEN TECH PLEDGE

1. Audit your group's technology habits for a typical week (devices, streaming, storage, AI use, charging).
2. Choose three personal actions and one action the school could take.
3. Estimate the impact of each action using the footprint cards from last lesson.
4. Create a "green computing pledge" poster for your classroom.

In your workbook, answer:

? Write your three personal actions and why you chose them.

? What action could the school take?

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is green computing?

A A type of virus
B Painting computers green
C Using technology in ways that reduce environmental harm
D Computers in gardens

2. What is heat reuse?

A Heating servers
B Reheating food
C Using data centre waste heat to warm buildings
D Overclocking a CPU

3. How does efficient code save energy?

A It does the job in fewer steps, using less processing
B It uses more servers
C It doesn't
D It's written in green

4. Which personal action has the biggest environmental benefit?

A Changing your wallpaper
B Keeping devices longer
C Buying a new phone yearly
D Using dark mode once

QUICK CHECK: ANSWERS

1. **C: Using technology in ways that reduce environmental harm**
   Sustainable tech.

2. **C: Using data centre waste heat to warm buildings**
   Turns waste into a resource.

3. **A: It does the job in fewer steps, using less processing**
   Fewer steps, less power.

4. **B: Keeping devices longer**
   Embodied carbon dominates.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three ways a data centre could reduce its environmental impact.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe green computing actions for data centres
- ✓ I can explain how efficient software reduces energy use
- ✓ I can make a personal green computing plan

END OF TOPIC

CLOUD COMPUTING AND GREEN COMPUTING

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.